Exercise Sheet 4

In all exercises $\{Z_t\} \sim \mathrm{WN}(\sigma^2)$.

1. Consider the model
$$X_t + 0.5X_{t-1} - 0.42X_{t-2} = Z_t$$
Discuss this model in detail and derive as many as possible properties we discussed in the course.

2. Take the MA(1) polynomial $\theta(z) = 1 - z$ and denote $\{X_t\}$ be the process and $\{Z_t\}$ the noise. Show that
$$E\Big(Z_t - \sum_{j=0}^{n}(1 - j/n)X_{t-j}\Big)^2 \to 0.$$
Discuss what this result means.

3. In Figure 1 we have plotted ACF and PACF two time series. Interpret the result and explain what model the could come from.

4. Consider the model
$$X_t - \phi X_{t-1} = Z_t + \theta Z_{t-1},$$
with $|\phi| < 1$ and $|\theta| < 1$. Drive the representation of X_t as a linear process.

5. In the last example derive $\gamma(h)$.

6. Apply the innovations algorithm to the process
$$X_t = \phi X_{t-1} + Z_t, \quad |\phi| < 1,$$

7. Apply the innovations algorithm to an MA(q) process and show that
$$\hat{X}_{n+1} = \sum_{j=1}^{q} \theta_{nj}(X_{n+1-j} - \hat{X}_{n+1-j}),$$
if $n \geq q$.

8. For an MA(1) process $X_t = Z_t + \theta Z_{t-1}$ we obtained by applying the innovations algorithm that
$$\hat{X}_{n+1} = \frac{\sigma^2\theta}{\nu_n}(X_n - \hat{X}_n),$$
with
$$\nu_n = \sigma^2\left(1 + \theta^2 - \frac{1}{\nu_{n-1}}\theta^2\sigma^2\right),$$
and $\nu_0 = \sigma^2(1 + \theta^2)$. Show that if $|\theta| < 1$ then $|\nu_n - \sigma^2| \leq C\rho^n$ for some $0 < \rho < 1$ and some large enough constant C (exponentially fast convergence). Also the convergence $\theta_{n1} \to \theta$ is exponentially fast.
What happens to ν_n if $|\theta| > 1$ or if $|\theta| = 1$?

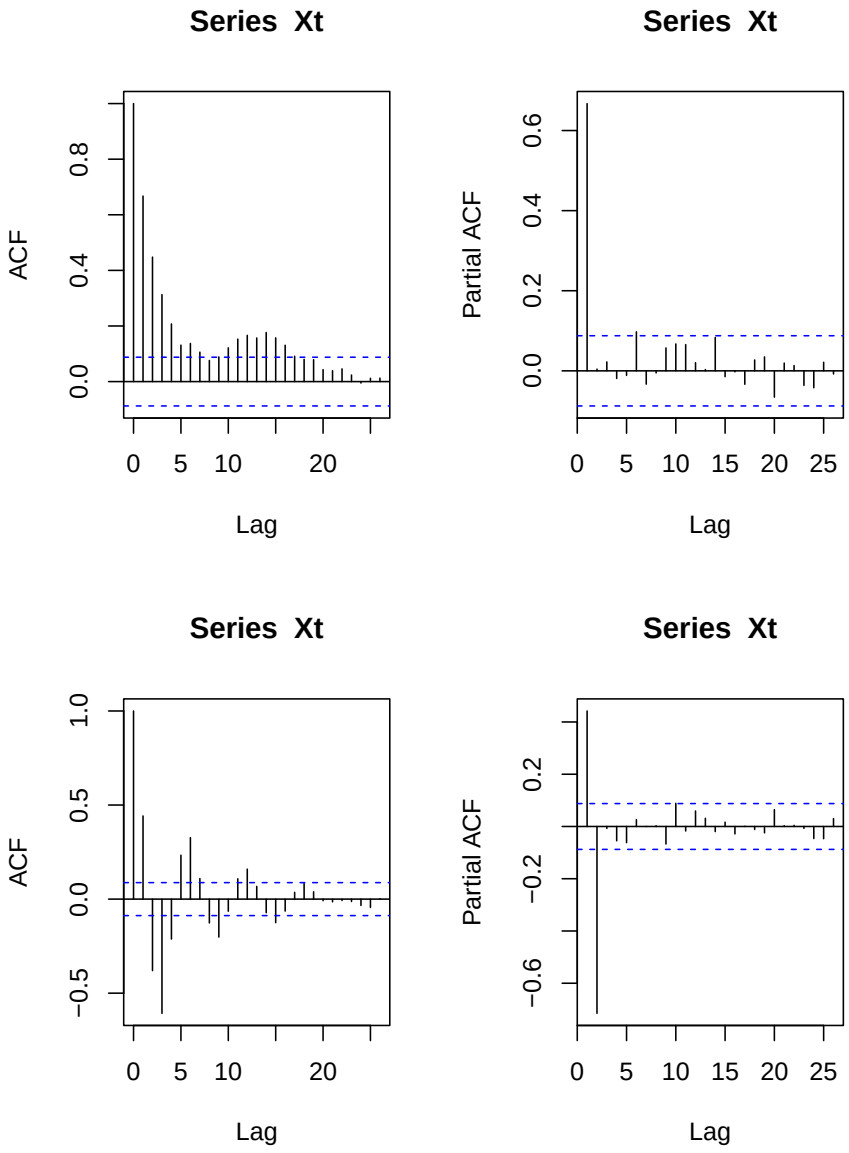


Figure 1: ACF and PACF of two time series.